



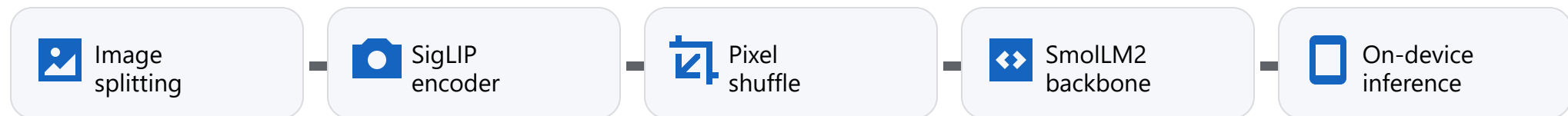
SmolVLM

Redefining Small and Efficient Multimodal Models

Compact VLMs for sub-1GB on-device inference

Hugging Face & Stanford University

Paper walkthrough • 2026-04-21



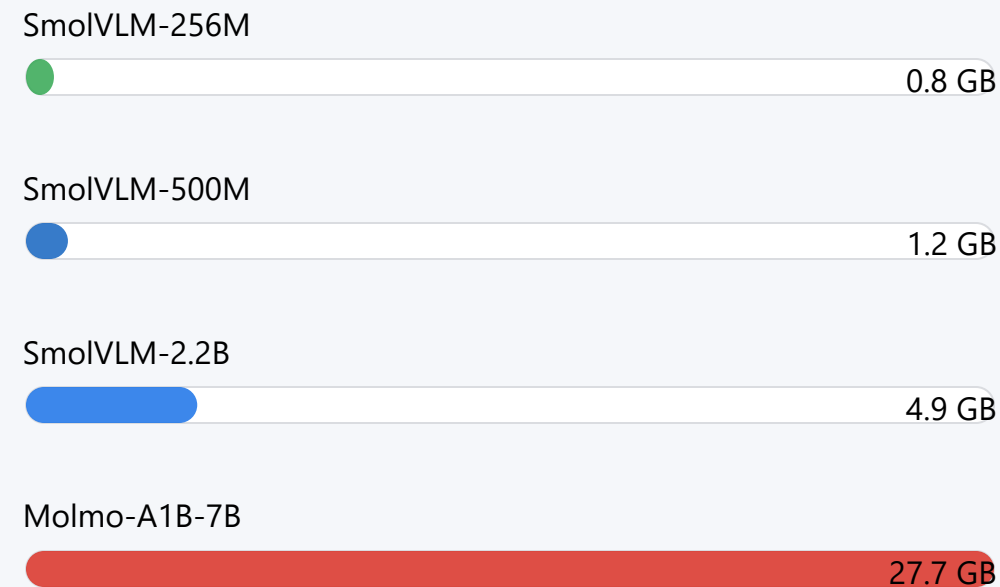
Large VLMs Are Powerful but Impractical for Edge Deployment

i Edge deployment is bottlenecked by memory; SmolVLM targets sub-1GB inference.

Edge Constraints

- ❗ Memory limits (RAM / unified memory)
- ❗ Power + thermal envelope
- ❗ Latency requirements (interactive UX)
- ❗ Privacy: on-device processing preferred

Memory Footprint Gap

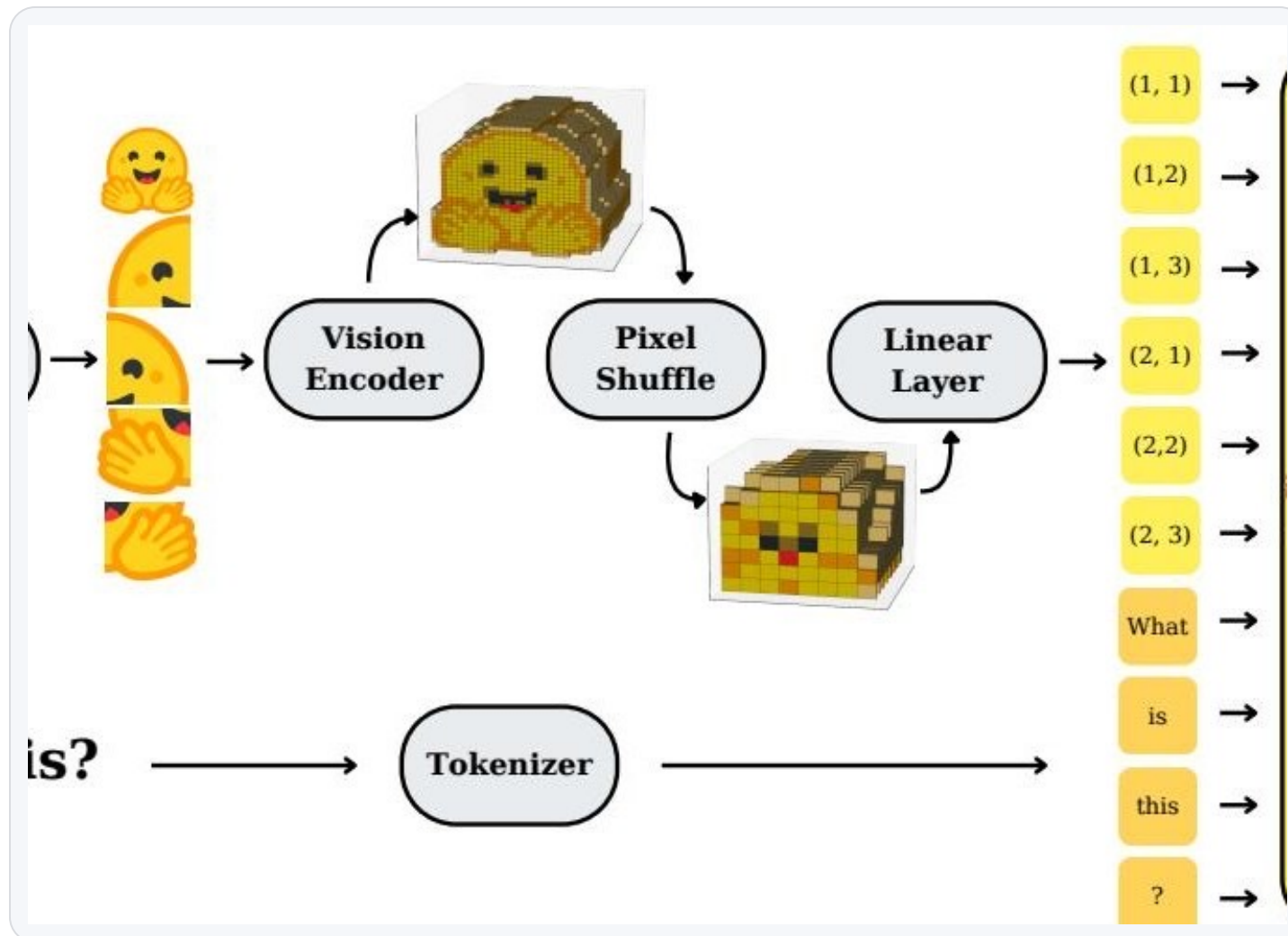


Target: sub-1GB inference for mobile/edge

SmolVLM-256M runs at 0.8 GB (batch=1).

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2 Backbone

i Pipeline: split image → SigLIP encode → pixel shuffle compress → project → SmolLM2.



Execution Flow (inference)

- ➔ 1) Image splitting
Split image into sub-images for higher effective resolution.
- ➔ 2) SigLIP vision encoder
Encode each sub-image into spatial feature maps.
- ➔ 3) Pixel shuffle (space-to-depth)
Reduce visual tokens by r^2 before attention.
- ➔ 4) Linear projection → SmolLM2
Project to LM embedding space and concatenate with text.

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

i For the smallest LMs, a 93M SigLIP-B/16 encoder beats a 428M SigLIP-SO400M encoder.

Small LM regime (135M LM): relative performance

SigLIP-B/16 (93M)



SigLIP-SO400M (428M)

Finding 1: balanced encoder–LM compute

93M encoder + 135M LM outperforms 428M encoder + 135M LM.

Only at 1.7B LM scale does the larger encoder justify ~10% overhead.

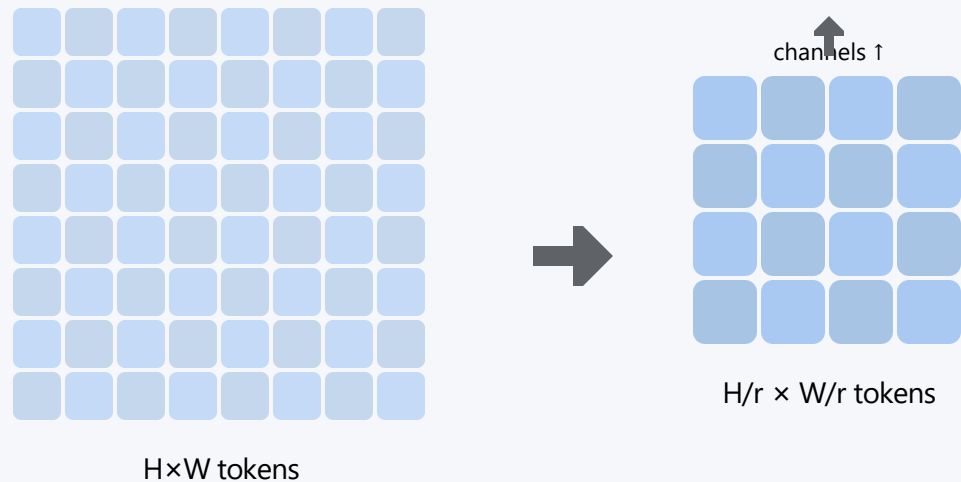
Implication for on-device VLMs

- ✓ In small LMs, attention/compute is
- ✓ Shrinking the encoder can reduce memory/latency without
- ✓ Allocate parameters where the model can use them

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models

i Pixel shuffle reduces tokens by r^2 ; $r=4$ yields 16× fewer visual tokens.

Mechanism: space-to-depth (pixel shuffle)



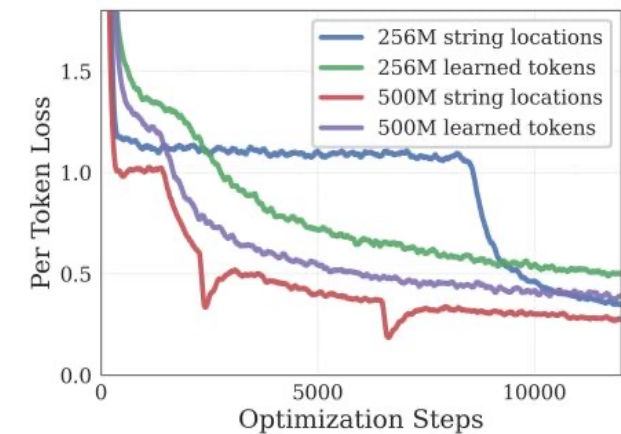
Token reduction

Visual tokens scale as $1/r^2 \rightarrow r=4$ gives 16× fewer tokens.

r=4: 16× reduction

Empirical note (from paper)

- ✓ Small VLMs benefit from $r=4$ compression.
- ✓ Larger models tend to prefer $r=2$.
- ✓ Compression eases attention overhead for



Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

 Longer context enables more visual tokens / higher resolution reasoning.

Context ladder

16K tokens

2.2B

8K tokens

256M/500M

2K tokens

How they scale context

 RoPE base update
Increase RoPE base from 10K to 273K.

 Mixed-context fine-tuning
Fine-tune on a mixture of long and short contexts.

 Deploy size-dependent context
2.2B uses 16K context; smaller variants use 8K.

 Observed effect
Performance improves significantly from 2K → 16K tokens.

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

 SmolVLM matches strong open baselines while cutting RAM by $>5\times$ – $30\times$.

Average across 13 benchmarks + RAM (batch=1)

Model	Params	Avg score	RAM	Efficient OS baseline
SmolVLM-256M	256M	44.0%	0.8 GB	Molmo-A1B-7B (27.7 GB RAM)
SmolVLM-500M	500M	51.0%	1.2 GB	Molmo-A1B-7B (27.7 GB RAM)
SmolVLM-2.2B	2.2B	59.8%	4.9 GB	InternVL2-2B (Efficient OS for video)

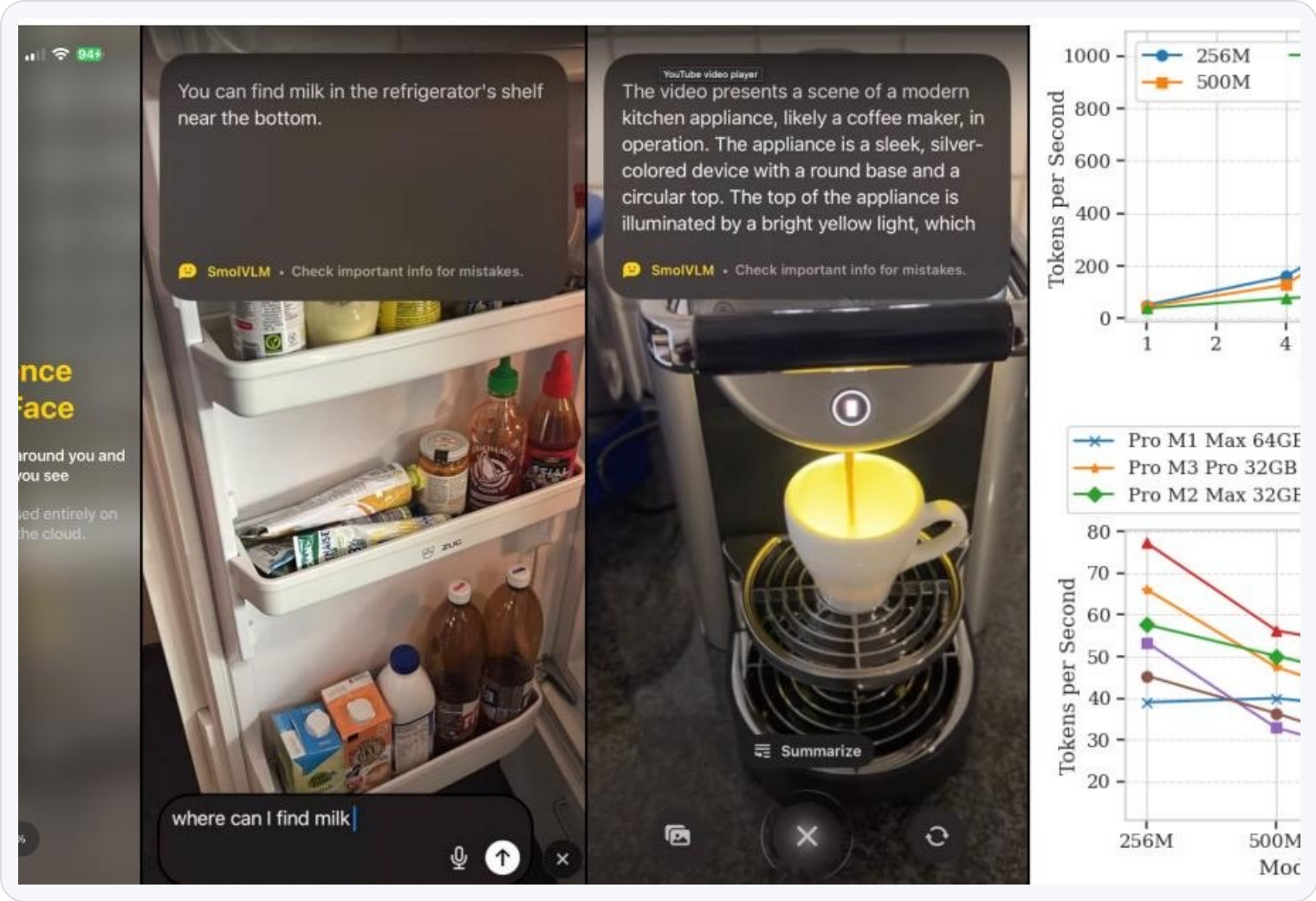


Memory contrast

Molmo-A1B-7B requires 27.7 GB RAM; SmolVLM-256M runs in 0.8 GB.
SmolVLM-256M also outperforms the $300\times$ larger Ldefics-80B (per paper).

Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon

i On-device throughput reaches ~80 tok/s for 256M, scaling with batch size.



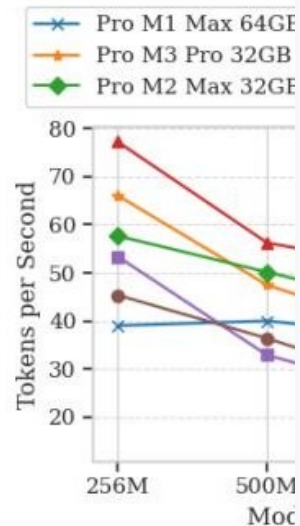
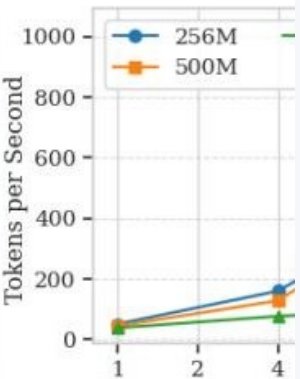
Deployment metrics (reported)

 ~80 tok/s
SmolVLM-256M on Pro M1 Max
64GB

 ~1000 tok/s
256M at batch size 64 on
high-end devices

 ~55 tok/s
256M on MacBook Air M3
16GB

 On-device
HuggingSnap: no cloud
dependency



Key Takeaways: Design Principles for Efficient Multimodal Models

 Architecture-first efficiency enables strong VLMs under tight RAM budgets.



Balance compute

Match encoder size to LM regime; small LMs favor smaller encoders.



Compress tokens

Pixel shuffle reduces visual tokens by r^2 ; $r=4$ helps small models.



Scale context

2K→16K via RoPE base 10K→273K + mixed-context fine-tuning.



Target sub-1GB

SmoVLM-256M runs at 0.8 GB RAM (batch=1).



Open by default

Weights, datasets, code, and HuggingSnap are open-sourced.

Implication: Efficient multimodal models are now practical on phones and edge accelerators.